

Signed last-layer normal form

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Lemma 1 (Signed last-layer normal form). *Let P and Q be probability laws on a finite subset of $\mathbb{Z}_{\geq 0}$, and assume that*

$$P(0) > 0.$$

Let

$$\{B_h\}_{|h| \leq n}$$

be a family of normalized signed measures of finite first moment, indexed by histories

$$h = (x_1, \dots, x_t), \quad x_i \in P.$$

Write

$$p(h) := \prod_{i=1}^{|h|} P(x_i), \quad p(\emptyset) := 1,$$

and define the local zero-repair residual

$$r_h := Q * B_h - \sum_x P(x) \delta_x * B_{hx}, \quad |h| < n.$$

Then every r_h has total mass zero.

Define the weighted prefix transport defect

$$\mathfrak{F}(B) := \sum_{1 \leq |h| < n} p(h) \|r_h\|_{\text{KR}},$$

where $\|\cdot\|_{\text{KR}}$ denotes the one-dimensional Kantorovich–Rubinstein norm on zero-mass signed measures.

Then there exists another family of normalized signed measures

$$\{\tilde{B}_h\}_{|h| \leq n}$$

with the same root law

$$\tilde{B}_{\emptyset} = B_{\emptyset},$$

such that

$$\boxed{\tilde{r}_h = 0 \quad \text{for every } 1 \leq |h| \leq n-2,} \tag{1}$$

and

$$\boxed{\sum_{|h|=n-1} p(h) \|\tilde{r}_h\|_{\text{KR}} \leq \mathfrak{F}(B).} \tag{2}$$

Thus, in the signed relaxation, every prefix defect can be pushed to the last independent prefix layer without increasing its total history-weighted Kantorovich–Rubinstein cost.

Proof. We first show how to eliminate one residual at one internal node.

Fix a history h with

$$1 \leq |h| \leq n - 2,$$

and put

$$p_0 := P(0) > 0.$$

Modify only the 0-child of h by setting

$$\boxed{B'_{h0} = B_{h0} + \frac{1}{p_0} r_h,} \quad (3)$$

while

$$B'_{hx} = B_{hx}, \quad x \neq 0.$$

Since r_h has zero total mass, B'_{h0} remains normalized. Moreover,

$$\begin{aligned} r'_h &= Q * B_h - \sum_x P(x) \delta_x * B'_{hx} \\ &= Q * B_h - \sum_x P(x) \delta_x * B_{hx} - p_0 \frac{1}{p_0} r_h \\ &= r_h - r_h \\ &= 0. \end{aligned}$$

Hence the residual at h has been removed exactly.

The only residual at the next layer which changes is the residual of the child $h0$. It becomes

$$\boxed{r'_{h0} = r_{h0} + \frac{1}{p_0} Q * r_h.} \quad (4)$$

Convolution by a probability measure is nonexpansive in the Kantorovich–Rubinstein norm:

$$\boxed{\|Q * \nu\|_{\text{KR}} \leq \|\nu\|_{\text{KR}}} \quad (5)$$

for every zero-mass signed measure ν of finite first moment. Indeed, by Kantorovich duality,

$$\|Q * \nu\|_{\text{KR}} = \sup_{(f) \leq 1} \langle f, Q * \nu \rangle = \sup_{(f) \leq 1} \langle Q^* f, \nu \rangle,$$

and Q^* preserves the 1-Lipschitz class.

Since

$$p(h0) = p(h)p_0,$$

the transported contribution of the newly created term satisfies

$$\begin{aligned} p(h0) \left\| \frac{1}{p_0} Q * r_h \right\|_{\text{KR}} &= p(h) \|Q * r_h\|_{\text{KR}} \\ &\leq p(h) \|r_h\|_{\text{KR}}. \end{aligned}$$

Consequently, by the triangle inequality,

$$p(h0) \|r'_{h0}\|_{\text{KR}} \leq p(h0) \|r_{h0}\|_{\text{KR}} + p(h) \|r_h\|_{\text{KR}}.$$

Thus eliminating the residual at h and pushing it one level downward does not increase the total weighted KR defect.

Apply this operation simultaneously to all histories of depth 1, then to all histories of depth 2, and continue successively through depth $n - 2$. Because the history graph is a tree, every child has a unique parent, and the preceding estimate can be iterated without ambiguity.

After the final step all residuals in depths

$$1, \dots, n - 2$$

vanish, while the entire remaining defect is supported on depth $n - 1$. At every pushing step the weighted KR cost is nonincreasing. Therefore

$$\sum_{|h|=n-1} p(h) \|\tilde{r}_h\|_{\text{KR}} \leq \sum_{1 \leq |h| < n} p(h) \|r_h\|_{\text{KR}}.$$

This proves (1) and (2). $\square$

[The only missing ingredient is positivity]

Lemma 1 is an exact statement in the space of normalized signed measures. The transformation

$$B_{h0} \mapsto B_{h0} + \frac{1}{P(0)} r_h$$

need not preserve coefficientwise nonnegativity.

Hence the algebraic and transport parts of the last-layer reduction are already complete. The remaining obstruction in the probability problem is precisely the existence of a positive realization of this nonexpansive signed normal form.

If

$$\mathfrak{F}_n^+$$

denotes the minimum weighted KR defect over the positive block-feasible face, and

$$\mathfrak{F}_{n,\text{last}}^+$$

denotes the corresponding minimum with the additional exact constraints

$$r_h = 0, \quad 1 \leq |h| \leq n - 2,$$

then

$$\mathfrak{F}_n^+ \leq \mathfrak{F}_{n,\text{last}}^+.$$

It is therefore natural to define the positivity premium

$$\Gamma_n := \mathfrak{F}_{n,\text{last}}^+ - \mathfrak{F}_n^+ \geq 0.$$

The signed lemma proves

$$\Gamma_n^{\text{signed}} = 0$$

identically. Thus any strict inequality

$$\Gamma_n > 0$$

in the original problem can arise only from the positive-cone constraint.

Conjecture 1 (Positive last-layer lifting). *For the quartic zero-repair pair*

$$P = \left(\frac{1}{8}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{8} \right), \quad Q = \left(\frac{7}{64}, \frac{5}{16}, \frac{5}{32}, \frac{5}{16}, \frac{7}{64} \right),$$

the positive block-optimal face admits a last-layer normal form with no additional KR cost:

$$\Gamma_n = 0.$$

A weaker scale-uniform statement,

$$\sup_n \Gamma_n < \infty,$$

would already isolate positivity as a bounded correction to the signed residual-pushing mechanism.